

Web 3.0 Unveiled - Day 2

Mentored by FEC

What is Ethereum?

Ethereum is a decentralized blockchain that supports smart contracts. Unlike Bitcoin, which only supports the transfer of the Bitcoin token around the network, Ethereum is more general purpose.

Developers can build dApps, or decentralized applications, which can be executed on the Ethereum network on the Ethereum Virtual Machine (EVM). Therefore, the global state of Ethereum consists of more than just the balance of every account but also the state of each dApp.

dApps are built on Ethereum using its programming language, Solidity. You can write smart contracts using Solidity and deploy the smart contracts to the Ethereum Network.

History

Ethereum was originally proposed by [Vitalik Buterin](#), a young programmer and co-founder of the Bitcoin Magazine. He was a curious student and a gifted programmer even at a young age. He used to play World of Warcraft from 2007-2010. However, when Blizzard decided to remove the damage component from his favorite warlock's Siphon Life skill, Buterin was very upset and quit the game altogether.

In 2011, he came across Bitcoin and got quite fascinated by the idea. He wanted to contribute formally, so he started writing articles about it in exchange for a few Bitcoin per article. In late 2011, he co-founded the Bitcoin Magazine and went full-time into crypto. He traveled across the world looking at various crypto projects. He realized it is possible to generalize the functionality of most of these projects by integrating a general-purpose programming language into a blockchain.

Buterin tried to convince the Bitcoin core developers, among devs from other projects, that blockchain technology can have a much wider impact than just monetary exchanges, and blockchain needed a general-purpose programming language to go with it. However, after being refused by the existing projects, he decided to do it himself. He proposed the development of a new blockchain platform with a programming language (Solidity), which went on to be what we know as Ethereum.

In 2014, the project was publicly announced, and the core team consisted of Vitalik Buterin, Mihai Alise, Anthony Di Iorio, Charles Hoskinson, Joe Lubin, and Gavin Wood. A few months later, the team held an [Initial Coin Offering \(ICO\)](#) for the Ether token to fund the development. The team raised over 31,000 BTC, which was around \$18 million USD at the time. They then established the Ethereum Foundation, a non-profit based in Switzerland, which was tasked with seeing Ethereum's open-source development early on.

What is Ether?

Ethereum has a native currency called "Ether" or "ETH". This token is required to pay transaction fees for transactions done on the Ethereum network.

What are Smart Contracts?

Smart contracts are small computer programs that are replicated and processed on all the computers on the Ethereum network without a central coordinator. Smart Contracts allow you to program contracts that computer code can automatically enforce.

The general-purpose nature of Ethereum allows for any number of possible applications to be built on top of it, which all inherit the security and decentralization benefits that come from running on the Ethereum blockchain.

ERC20 Tokens

In addition to Ether, people can create and use their own currencies on Ethereum. The most common form of currency is ERC20 tokens. **ERC20 Tokens are smart contracts that fit a specific standard.** Developers can extend beyond the standard but should **meet the minimum requirements** when making their own token. The standardization allows digital wallets to easily support all types of tokens without needing specialized code for each token created.

ERC721 and ERC1155 Tokens

These are what are also called **NFTs**. These two standards, similar to ERC20, provide a baseline for what requirements should be met when creating NFTs. They also provide similar benefits, allowing wallets and NFT marketplaces to instantly be compatible with all NFT collections as they all follow one of these standards.

Must Watch:  [NFT Explained In 5 Minutes | What Is NFT? - Non Fungible Token | NFT Crypto E...](#)

Additional Reading (Optional):

- [ERC-20 Tokens, Explained - Cointelegraph](#)
- [Smart Contracts - Simply Explained](#)
- [What Are ERC-20 Tokens on the Ethereum Network?](#)

Setting Up a Crypto Wallet

Introduction

To understand crypto wallets fully, we have to understand some concepts about the blockchain, which will help us understand how a wallet aids us. Let's start off.

What is an address?

An address is a string of text generated using cryptography to represent your account on the blockchain. This address can be shared publicly with others, and it is completely safe to do so. You can send and receive funds from and to your wallet address. Basically, the address is your unique identifier on the blockchain and represents your 'account.' An example of an Ethereum address is: `0x01573Df433484fCBe6325a0c6E051Dc62Ab107D1`.

What are private keys?

A private key is the counterpart to an address. Each address has an associated private key. As the name suggests, though, this is meant to be kept private and not shared with anyone.

You can think of it like a password—a really strong one that contains a bunch of letters and numbers that allow you to prove ownership over your address. Anyone who has the private key has access to make transactions from your address, i.e., send money from your address to theirs.

A private key looks something like this:

`E9873D79C6D87DC0FB6A5778633389F4453213303DA61F20BD67FC233AA33262`.

If you think of your address as a username for your account, the private key is its password. Therefore, sharing your address is okay, but never share your private key, or someone might steal your funds - and nothing can be done about it.

Caution: Since blockchains are decentralized, there is no 'forgot password' option. If you lose your private key, you lose access to your account. Similarly, if someone steals your private key and steals your funds, you cannot do anything about it. It is VERY important to keep this private key safe.

For developers, we often use the private key as part of our codebase to perform certain transactions, such as deploying our own smart contracts to the Ethereum network. While you are still learning, we highly suggest you use a separate account entirely for development than you use for storing any sort of funds. Unfortunately, beginner developers often use the same

account they have funds on and accidentally share their codebase publicly - and hackers can see your private key in the codebase and end up stealing funds. Please take that as a tale of caution.

Or you can watch this video: [▶ How Public and Private Key Work In Your Crypto Wallets](#)

What is a seed phrase?

A seed phrase is like a master password - the password of passwords!

Think of a password manager, something like Lastpass or 1Password. These applications, within them, store your usernames and passwords for other apps securely, and themselves have a password. So, if someone hacks your password manager, they also get access to all accounts stored within it.

A crypto wallet is kind of like a password manager, where you can manage multiple blockchain accounts. If the private key is the password to a single account, the seed phrase is kind of like the master password for that wallet.

When you create a new crypto wallet, you will be presented with a seed phrase you should absolutely securely store and back up. Any new accounts you generate from inside that wallet will all be linked to the seed phrase. That one seed phrase will always generate the same accounts, with the same private keys and addresses for each.

So, for example, if you created a wallet and then created 5 accounts within it, your seed phrase manages all 5. If you wanted to switch to a new wallet, you could either import the 5 wallets individually - by using their individual private keys - or just import using the seed phrase, and it would regenerate the same 5 accounts.

An example of a seed phrase is: `dove lumber quote board young robust kit invite
plastic regular skull history`

Or you can watch this video:

[▶ Crypto Seed Phrases VS Private Keys: What's The DIFFERENCE??](#)

What is a crypto wallet then?

Crypto wallets are a manager for your accounts and mainly their private keys. They also allow you to interact with decentralized applications and connect to a dApp through the wallet, acting as a single sign-on for all applications built on the blockchain.

Setting up a Wallet

For Ethereum, there are a number of wallet options available. The easiest to get started using, and most developer friendly, are either MetaMask or Coinbase Wallet.

Both are Ethereum crypto wallets that can be installed as browser extensions or as mobile apps. We will download MetaMask for this course. You can find the download link and tutorial below.

- [Download MetaMask](#)
- [MetaMask tutorial](#)